Design and Validation of a Pilot Stress Monitoring System based on a Deep Convolutional Neural Network Model

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***Abstract* - This article presents the Design and Validation of a Stress Model based on a Deep Convolutional Neural Network (DCNN) included in the Pilot Monitoring System (PMS) developed by Airbus Defence Space.**

Starting by introducing the whole system, PMS obtains pilot's vital data in real time with two wearable devices (smartwatch and ear sensor). It is able to determine stress, fatigue, workload, health status and pilot capacity with new models based on DCNN encoded in Python which are able to interpret these vital data.

Additionally, PMS is able to launch alerts when the vital parameters of the crew reach pre-established limits, giving the chance to detect an abnormal situation (e.g. excessive workload scenario with high stress level). In this way, it will improve situational awareness by making crew aware of their own condition

Going deeper into the details of the Stress Model, its main objective is to determine Pilot stress in real time.

For achieving this target, several design stages were taken into consideration while building the Stress Model DCNN: raw data acquisition, data cleaning, feature engineering, algorithm selection, model training, model evaluation, model testing, and result analysis. Further details of these stages are explained below.

Regarding data acquisition, Stress Model obtains pilot's vital data from a smartwatch (one of the wearable devices of PMS), which communicates via Bluetooth Low Energy (BLE) to a laptop. Selected device was Analog Devices Inc. Gen II watch, which sends the following signals: Heart Rate (HR), Electro-Dermal Activity (EDA), Skin temperature and 3D Acceleration.

At Cleaning stage, Inter-Quartile Range (IQR) was used to normalize our raw data, being less sensitive to outliers and therefore more robust.

At Feature engineering stage, several signal combinations were explored. The selected signal set which gave the best Stress Model accuracy was comprised of two EDA parameters (Real Admittance [Siemens], Impedance Magnitude [Ohm]) and wrist Skin Temperature [deg C].

At Algorithm selection stage, three convolutional layers (128, 256 and 128 filters respectively) were chosen to extract a feature map from the inputs. After them, a Global Average Pooling (GAP) was chosen to compress the convolutions output. Downstream GAP, a decision layer based on softmax activation function (stress vs non-stress) was chosen.

At Training stage, it was necessary to implement a Stress Test recording campaign. Participants were asked to fulfill several individual demanding tasks (memory, multi-task, stroop, vigilance) which included relaxing periods among them. The Stress Model was fed with the vital data obtained in this campaign.

In between the tasks, NASA-TLX questionnaires were filled by test subjects, to obtain their subjective rating of Mental demand, Physical demand, Temporal demand, Performance, Effort and Frustration. Therefore, the Stress Model considered both vital data and self-report questionnaires as inputs.

At result analysis stage, the Stress Model accuracy was refined, understanding accuracy as the ratio between correct predicted stress scenarios versus total cases, and considering ground truth self-report questionnaires.

Finally, Stress Model DCNN was validated considering three data sources, whose details are included below:

A. Vital data from fifteen (N=15) subjects of Wearable Stress and Affect Detection (WESAD) Database available online.
Stress Model gave an average test accuracy of 99%,

B. Vital data from four (N=4) Airbus Defence and Space (ADS) pilots while calibrating their NASA-TLX scale.
Stress Model in this validation gave an average test accuracy of 90%.

C. Vital data from six (N=6) ADS pilots while performing simulator evaluations using an Enhanced Light Weight Visor (ELWEV).
Stress Model in this validation gave an average test accuracy of 87%.

Keywords— Pilot Stress, Deep Convolutional Neural Network, Electro-Dermal Activity, NASA-TLX, wearable devices

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I. INTRODUCTION

This report presents the Design and Validation of a Stress Model based on a Deep Convolutional Neural Network (DCNN) included in the Pilot Monitoring System (PMS).

In this framework, Airbus Defence Space (ADS) is developing the Pilot Monitoring System (PMS) including a Stress Model.

Starting by introducing the whole system, PMS obtains pilot's vital data in real time with two wearable devices (smartwatch and ear sensor). It is able to determine stress, fatigue, workload, health and pilot capacity based on new algorithms based on DCNN encoded in Python which are able to interpret these vital data.

Additionally, PMS is able to launch alerts (shown in Fig. 1) when the vital parameters of the crew reach the pre-established limits, giving the chance to detect abnormal situation (e.g. excessive workload scenario with high stress level). In this way, it will improve situational awareness by making crew aware of their own condition

The personal stress, fatigue, workload, health and capacity data can be stored in a database. This functionality could improve the fleet management by making a more rational distribution of the workload between the different crewmembers to equilibrate the efforts.

Going deeper into the details of the Stress Model, its main objective is to determine Pilot stress in real time. To achieve this, it is needed to acquire vital data from them.

As described in [1], the most interesting vital signals historically related to stress are: Blood Volume Pressure (BVP), Galvanic Skin Response (GSR) or Electro Dermal Activity (EDA), Pupil Dilatation (PD), Skin Temperature (ST), Electrocardiogram (ECG), Breathing Rate (RR) and Electromyogram (EMG).

More recent studies, as [2], consider EDA is a key signal to the Stress Detection.

Therefore, PMS has integrated ADI Gen II smartwatch capable of acquiring EDA from pilots (among other parameters such as Skin temperature, 3D Acceleration through ADXL sensor).

Fig. 1. Example of alerts launched by PMS

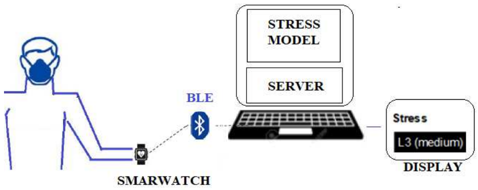

Fig. 2. System architecture

As shown in Fig.2, the signals from smartwatch are sent to a computer, via Bluetooth Low Energy (BLE), where they are then prepared for training or directly feed the Stress Model encoded in Python passing by a server.

II. METHODS

In this section, more details are given regarding the Stress Test Campaign, the Stress Model DCNN design and the Stress Model validation campaigns.

A. Stress Test Campaign Participants and Study Protocol

A Stress Test recording campaign was implemented in order to obtain vital data in a controlled environment.

19 healthy ADS volunteers (31.3 ± 7.8 years, age range 24 – 50 years, 21% women) were recorded during the Stress Test campaign mentioned above.

The participants selected for this Stress Campaign had no medical contraindications such as severe concomitant disease, alcoholism, psychological or intellectual problems which could limit compliance into the study.

A safety-hygiene protocol has been strictly followed to keep on testing after Covid-19 pandemic.

As part of this campaign, participants were asked to fulfill several demanding tasks (memory, multi-task, stroop, vigilance) which included relaxing periods among them. The complete test sequence is shown in Table 2

Table 1. Stress test campaign tasks sequence

Task	Time	Comment
N-back (1) Test	2min	Memory
NASA-TLX	1min	
Fatigue Questionnaire	5min	KSS
Resting State	5min	Video
N-back (2) Test	3min	Memory
NASA-TLX	1min	
N-back (3) Test	3min	Memory
NASA-TLX	1min	
Resting State	2min	Video
Multi-task Test	5 min	Multitasking
NASA-TLX	1min	
Resting State	2min	Video
Stroop Test	8min	Interference Inhibition
NASA-TLX	1min	
Vigilance Test	5min	Sustained Attention
NASA-TLX	1min	
Fatigue Questionnaire	5min	KSS

To better illustrate the Stress Test tasks, a short description is included below.

N-back (1) test stimulated the memory skill of the participant. A letter was presented in the screen, and it changed after a time step. The task consisted in comparing a letter in the screen with the previous one shown in the screen.

N-back (2) test also stimulated the memory skill, but with a higher difficulty, because the letter shown in the screen must be compared with the letter shown in the screen two time steps ago.

N-back (3) stimulated analogously the memory skill, with the maximum difficulty because it required to compare the letter shown in the screen with the letter shown three time-steps ago.

Multitask test stimulated the ability to manage conflicting information provided by the direction of arrows and its location on the screen and to ignore task irrelevant information.

Stroop test was a demonstration of interference in the reaction. It permits to measure a person selective attention capacity and skills, by mixing font letters and words meaning, e.g. 'red' written with green font.

Ending the Stress Test Campaign, vigilance task stimulated reaction time in a monotonous task, which was observing the advance of a clock's second hand, identifying when an anomalous time-jump appeared.

At resting state periods, different relaxing videos were played in order to recover from stress state induced by the rest of the tasks.

On the other hand, NASA-TLX questionnaires were filled by test subjects in between the tasks.

NASA Task Load Index (NASA-TLX) is a subjective methodology used to assess test subjects perceived workload when performing a task. NASA-TLX (described in [3], [4], [5]) questionnaires were filled by test subjects after each task to obtain their subjective rating of six dimensions: Mental demand, Physical demand, Temporal demand, Performance, Effort and Frustration. These self-report questionnaires filled by subjects were considered as ground truth.

An example of NASA-TLX questionnaire is depicted in Fig. 3.

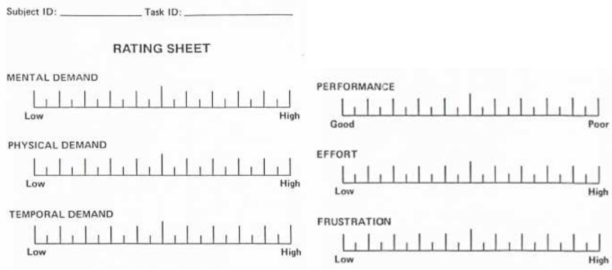

Fig. 3. NASA-TLX questionnaire

Finally, fatigue questionnaires were also filled by test subject, including several subjective scales (fatigue, sleepiness, activation) and Karolinska Sleepiness Scale (KSS).

B. Stress Model Development

Several design stages were taken into consideration while building the Stress Model DCNN: raw data acquisition, data cleaning and normalization, feature engineering, algorithm selection, model training, model evaluation, model testing, and result analysis, as shown in Fig. 4.

Further details of these design stages are explained below.

Regarding data acquisition, the raw data recorded in the Stress Campaign using ADI smartwatch was transferred to the development computer through BLE (as depicted in Fig. 2) and once there pre-processed.

At Pre-processing or cleaning stage, several steps were included. The first step was aligning the time stamps of the different sensors within the smart device (EDA, PPG, ST, ADXL). The second step was extracting the relevant data aligned with the time stamps of the different tasks presented in Table 2. Finally, the third step was normalizing the raw data by using Inter-Quartile Range (IQR). In this way, the model is less sensitive to outliers and therefore more robust.

As per [6], IQR is the range between the 1st quartile (25th quantile) and the 3rd quartile (75th quantile).

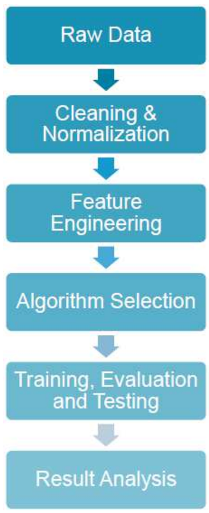

Fig. 4. Stress Model Development stages

Table 2: Feature Exploration

Signal Input Combination	**Accuracy in Test**
admreal(s)(EDA)	93%
ntc2 (ST)	89%
impmagnitude [Ω] (EDA)	87%
admreal(s)(EDA) + ntc2(ST)	97.4%
admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA)	**97.8%**
admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA) + EDA_tonic	97.9%
admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA) + EDA_tonic + EDA_phasic	98.2%
admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA) + EDA_tonic + EDA_phasic + EDA_smna	80%
admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA) + EDA_tonic + EDA_smna	74%
hr	70%
hr + admreal(s)(EDA)	92%
hr + admreal(s)(EDA) + ntc2(ST)	97.8%
hr + admreal(s)(EDA) + ntc2(ST) + impmag[Ω](EDA)	96.5%

During Feature engineering stage, the state-of-the-art literature was reviewed ([1], [2], [7], [8], [9]), [10], [11], [12]) and a battery of tests was done (see Table 3), obtaining a signal combination trade-off.

The selected signal set which gave the best Stress Model accuracy while maintaining an appropriate time and computation performance was comprised of two EDA parameters (Real Admittance [S], Impedance Magnitude [Ω]) and wrist Skin Temperature [deg C].

At Algorithm selection stage, we based our first approach on [2]. In it, there were different optimal architectures for Stress Detection depending on the input selected: Fully Convolutional Network (FCN) for Skin Temperature (ST) and Blood Volume Pressure (BVP), and Multi-Layer Perceptron (MLP) for voice. In case of combining multiple inputs, the proposed approach was to concatenate the previous architectures using a concatenation layer and then using a standard transformation to obtain the output.

Taking this information into account, and given that the signal inputs from our smartwatch were being cleaned and normalized, the network in Fig. 5 is proposed. The network singularity resides in the way inputs are ingested, in order to optimize processing performance and accuracy.

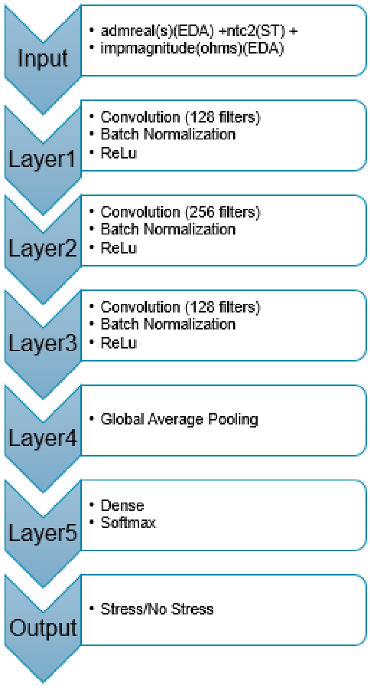

Fig. 5. Multi Input Fully Deep Convolutional Neural Network

The strength behind this architecture relies in the 3 convolutional layers (128, 256 and 128 filters respectively). Convolutional layers, frequently used in image processing, are able to extract a feature map from the inputs provided to them. The feature maps maintain the most important information while reducing the dimensionality of the data through training, so that the objective of optimizing processing performance is fulfilled.

After these three layers, a Global Average Pooling (GAP) is introduced. It is the most suitable solution to compress the information that the convolutions output. Finally, the output from GAP passes to the decision layer that will give the desired output through a softmax activation function: “stress” or “not stress”.

At Training stage, the approach depicted in Fig. 6 was applied for training Stress Models. The data of each pilot was divided into 2-second windows and randomly divided into 3 sets, 60% of data for ‘Training’, 20% for ‘Evaluation’ and 20% for ‘Testing’.

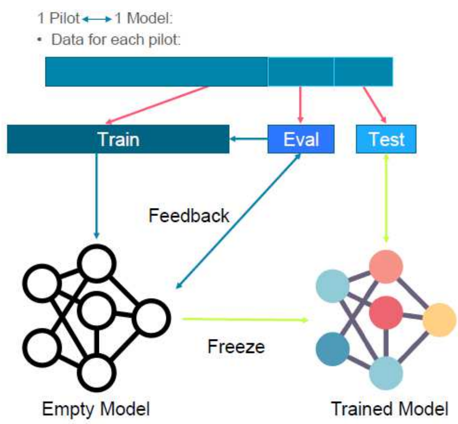

Fig. 6. Training strategy.

C. Stress Model Validation

In the end, Stress Model was validated considering three data sources, whose details are included below.

- Validation with WESAD Database

WESAD database is a publicly available multimodal dataset [13], with physiological signals and motion data, recorded from both a wrist- and a chest-worn device, of fifteen subjects during a lab study.

The Stress Model presented in this article gave an average test accuracy of 99%, being the ratio between correct predicted stress scenarios versus total cases, and considering ground truth self-report questionnaires filled by test subjects.

- Validation with ADS Pilots NASA-TLX calibration Database

NASA-TLX calibration makes use of the Multi-Attribute Task Battery (MATB-II) tool, consisting of four scenarios following an incremental approach. Each scenario includes all tasks performed in previous scenarios and an additional task, consisting in a total of four different tasks: tracking, system monitoring, resources management and communication. MATB-II interface is shown in Fig. 7.

Each of these tasks represent tasks performed by pilots during operation and are performed using a set-up which consisted of a laptop, a mouse, a joystick and a headset. NASA-TLX questionnaires were filled by pilots after each scenario. These self-report questionnaires were considered as ground truth. Further details about NASA-TLX questionnaires are available on [5].

Stress Model in this validation gave an average test accuracy of 90%.

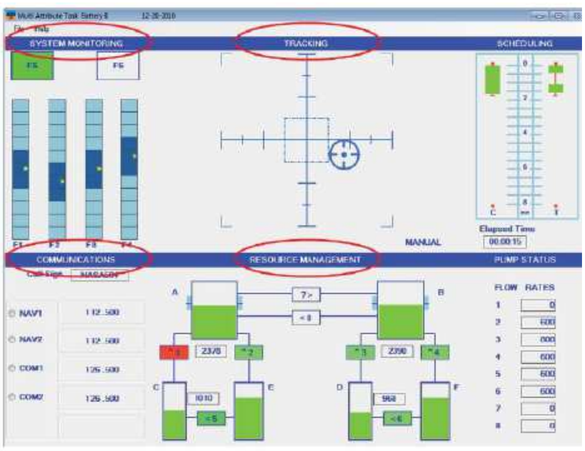

Fig. 7. Example of MATB-II Interface

- Validation with Pilot Vital Data while evaluating an Enhanced Light Weight Visor (ELWEV) visor

ELWEV testing campaign consisted on 3 different scenarios which aimed at representing real situations of regular aircraft operations, each performed with and without the ELWEV system: a low visibility taxi, a low visibility take-off and an in-flight emergency procedure execution after a failure. A total of 6 scenarios per pilot were registered, and NASA-TLX questionnaires were filled by the subjects after each scenario. These self-report questionnaires were considered as ground truth.

Stress Model in this validation gave an average test accuracy of 87%.

III. CONCLUSIONS

The results of the Stress Model, based on a multiple input DCNN, on the different data sets are summarized in Table 4.

Laboratory results on WESAD and Stress campaign look promising with State of the Art performances of 99% and 98% accuracies respectively. These results clearly demonstrate the potential of EDA and ST as indicators of stress in humans.

Laboratory results were not fully representative of the cockpit environment. This is the reason behind testing the Stress Model in ADS CS2 A330 Active Cockpit simulator with real pilots.

During the tests in simulator, an accuracy drop of 10% was observed. This drop was expected, and the performance of the model is still considered acceptable.

IV. FUTURE WORKS

Next steps to be followed with Stress Model will include
-Full evaluation of PMS, validating the Stress Model with the rest of models (Fatigue, Workload, Health, Pilot Capacity) in Active Cockpit simulator
-Obtaining clearance for flight.
-In-flight validation (stand-alone configuration) recording Pilot-Data
-Model tuning with vital data obtained in flight for a given Pilot population.
-System upgrade, including a smartwatch capable of measuring Oxygen saturation (SpO_2). E.g. Analog Devices is validating SpO_2 algorithm for Gen IV watch.

Table 3: Results of the DFCNN model over different data sets

Dataset	Accuracy
Stress Campaign 19 ADS subjects	98%
ADS Pilots NASA-TLX Calibration	90%
Pilot Vital Data while evaluating an Enhanced Light Weight Visor (ELWEV)	87%
WESAD	99%

V. ACKNOWLEDGEMENT

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VII. DISCLAIMER

The results, opinions, conclusions, etc. presented in this work are those of the author(s) only and do not necessarily represent the position of the JU; the JU is not responsible for any use made of the information contained herein.

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